ELSEVIER

New Astronomy 3 (1998) 27–36

NEW ASTRONOMY

Primordial LiH: the chemistry in a collapsing protocloud

Denis Puy[a,b,1], Monique Signore[c,2]

[a]*Paul Scherrer Institute, Laboratory for Astrophysics, CH-5232 Villigen PSI, Switzerland*
[b]*Institute of Theoretical Physics, University of Zurich, Winterthurerstrasse 190, CH-8057 Zurich, Switzerland*
[c]*Observatoire de Paris, Département de Radioastronomie Millimétrique, 61, Avenue de l'Observatoire, 75014 Paris, France*

Received 11 July 1997; accepted 9 October 1997
Communicated by Francesco Melchiorri

Abstract

We analyse and discuss the evolution of primordial molecules during the beginning stage of the early cloud formation using an updated primordial lithium chemistry. As Puy & Signore (1996)[A&A, 305, 371], (1997)[NewA, 2, 299], we consider a simple model of a collapsing protocloud after the recombination epoch but paying special attention to the LiH formation by radiative association from excited Li atoms. Because the rate of LiH destruction through collisions with H atoms is only estimated we find that, at present, *the percentage of primordial lithium converted to LiH is quite uncertain.*
© 1998 Elsevier Science B.V.

PACS: 95.30.Ft; 98.62.Ai
Keywords: Early universe; Molecular processes

1. Introduction

This paper deals with the chemistry of LiH in a model of a collapsing single isolated cosmological protocloud after the recombination epoch in an expanding Universe. In a first paper (Puy & Signore, 1996), we have presented the basic processes of physics and chemistry, and in a second one (Puy & Signore, 1997) we have analysed the thermal evolution and the influence of primordial molecules on the thermal history at the beginning of the collapse.

Primordial molecules can play a role in observational cosmology, and more precisely the study of lithium and his chemical component, LiH molecules, can offer interesting perspectives for Cosmology (see for instance, Signore et al., 1994).

In the framework of the standard Big Bang model, the nucleosynthesis (SBBN) of the light elements took place at $t_{nuc} \sim 10^2$ sec after the Big Bang. Cosmologists compute and predict the abundances of deuterium, helium-3, helium-4 and lithium-7 as a function of the ratio of nucleus to photons η and

[1]E-mail: puy@physik.unizh.ch
[2]E-mail: signore@mesiob.obspm.fr

1384-1076/98/$19.00 © 1998 Elsevier Science B.V. All rights reserved.
PII S1384-1076(97)00038-9

predict also very tiny abundances of heavier elements, η or equivalently Ω_b, the density of baryons in units of the critical density, is the SBBN free parameter. In particular (Sarkar, 1996):

$$\left(\frac{^7\mathrm{Li}}{\mathrm{H}}\right)_{\mathrm{SBBN}} = 1.2 \times 10^{-11\pm0.2} \langle\eta\rangle$$

where

$$\langle\eta\rangle = \left[\left(\frac{\eta}{5.\times 10^{-10}}\right)^{-2.38} + 21.7\left(\frac{\eta}{5.\times 10^{-10}}\right)^{+2.38}\right]$$

over the range: $\eta = 10^{-10}$–10^{-9}. Although the comparison of the predicted elemental abundances with observational data is complicated by the fact that the primordial abundances may have been significantly altered during the lifetime of the Universe by, in particular, nuclear processing in stars, astrophysicists identify the primordial ^{7}Li abundance with the observed average value in the hottest most metal poor stars:

$$\left(\frac{^7\mathrm{Li}}{\mathrm{H}}\right)_{\mathrm{pop.II}} = 10^{-9.78\pm0.2} \quad (92\%\ \mathrm{cl})$$

Moreover, the observation that:

$$\left(\frac{^7\mathrm{Li}}{\mathrm{H}}\right)_{\mathrm{pop.I}} \sim 10^{-9}$$

in young population I stars then requires the galactic (^{7}Li/H) ratio to rise by a factor of about 10 during the evolution of the Galaxy.

In the framework of the inhomogeneous nucleosynthesis Big Bang models (IBBN), the nucleosynthesis occurs in an homogeneous medium, due to fluctuations generated by a first-order quark hadron phase transitions at $T_{\mathrm{QH}} \sim 150$–400 MeV, i.e. at $t_{\mathrm{QH}} \sim 10^{-5}$ sec after the Big Bang. These inhomogeneities may have created neutron-rich zones where extra lithium-7 and heavier elements may have been formed. In particular, in this opposite point of view, the primordial ^{7}Li abundance may have been at the level of that observed in young population I stars, if, it has been depleted in the older population II stars, through some turbulent mixing driven by stellar rotation. For recent reviews on the status of the big bang nucleosynthesis (SBBN and IBBN) see Copi et al. (1995), Sarkar (1996), Schramm & Turner (1997) and references therein. Let us only note that observations of primordial level of the ^{7}Li nuclei would help to discriminate between SBBN and IBBN models and therefore to determine what our Universe looked like 10^{-5} sec after the Big Bang.

In standard Big Bang Theory, chemistry took place around the epoch of recombination. Many researchers have studied the evolution of the primordial LiH molecule (Lepp & Shull, 1984; Dalgarno & Lepp, 1987; Latter & Black, 1991), have searched for it (De Bernardis et al., 1993) and have shown that thanks to its possible prominence one can have a tool to analyse the ^{7}Li abundance at a progalactic level (De Bernardis et al., 1994; Melchiorri & Melchiorri, 1994; Maoli et al., 1994, 1996; Signore et al., 1994, 1997). In these papers, they have analyzed in detail, the main molecular radiation processes between CBR photons and primordial molecules in the context of two cosmological environments, on the one hand during the expansion of the Universe and on the oher during the gravitational collapse phase of the structures of the Universe. In particular, they have determined the conditions for blurring of primary CBA (cosmic background anisotropies) and for creation of secondary CBA. In fact Dubrovich (1993) suggested that the presence of molecules, particularly LiH molecules, can create a *curtain* that blurs the primordial spatial fluctuations of the cosmic background radiation (CBR), and moreover imprint on CBR from the resonant enhancements of the Thomson scattering process. In this context Signore et al. (1997) pointed out that maps made by the PLANCK SURVEYOR mission (formerly referred to as the COBRA-SAMBA mission, 1996) – the major space mission of the next decade dedicated to the CBR – should be excellent tools for discovering and analyzing anisotropies from resonant scattering of primordial molecules with CBR photons.

In summary one can say that primordial LiH signals are important not only as diagnostic tools for primary nucleosynthesis models and therefore for information on the physical conditions of the Uni-

verse at some 10^{-5} seconds after the Big Bang but also as tracers of protostructure later in the evolution of the Universe. In order to predict the observability of the LiH lines at high redshifts – or more generally of the effects of LiH molecules on CBR – let us consider the evolution of our new lithium chemistry and therefore let us estimate the final LiH abundance during the gravitational collapse of protostructure.

The chemistry of the primordial gas after the recombination epoch has been studied by several groups (Lepp & Shull, 1984; Dalgarno & Lepp, 1987; Latter & Black, 1991; Puy et al., 1993). More recently Stancil et al. (1996) have developed a new model of lithium chemistry in the early Universe. They include, in particular, new quantal rate coefficients for the radiative association of lithium and hydrogen (Dalgarno et al., 1996).

In this paper, we propose to extend the previous study in our simple model of a collapsing protocloud, and to include the new chemistry of Stancil et al. (1996) and additional results obtained by Gianturco & Gori Giorgi (1996) on the radiative association of LiH from electronically excited lithium atoms. In Section 2, we describe the new lithium chemistry in the early Universe; In Section 3, we present the evolution of the new lithium chemistry during the gravitational collapse of a protostructure, then we discuss the results and give conclusions.

2. The lithium chemistry in the early Universe

In Standard Big Bang theory, primordial chemistry took place around the epoch of recombination. At this stage, the chemical species essentially were H, H^+, D, D^+, He and Li (Lepp & Shull, 1984; Dalgarno & Lepp, 1987; Latter & Black, 1991; Puy et al., 1993). Then with the expansion and adiabatic cooling of the Universe, different routes led to molecular formation.

In spite of the absence of dipole moment for H_2 and the slowness of direct radiative association of two atoms (there are no grains), Shchekinov & Entel (1983) and Saslaw & Zipoy (1967) brought out two ways of formation for H_2 molecules through associative detachment with H^- and by the creation of H_2^+ by radiative association, followed by charge transfer. Lepp & Shull (1984) and Puy et al. (1993) shown that HD molecules mainly form through collision with H_2 molecules. The situation is very different concerning LiH molecules. Lepp & Shull (1984) suggested that the formation was provided by the only reaction:

$$Li + H \mapsto LiH + h\nu \quad \text{(radiative association, 28)}$$

Knowing that the ionisation potential of Li(I) is shorter than that of hydrogen (Li(I) = 5.392 eV), Dalgarno & Lepp (1987) suggested the existence of Li^+ ions after the recombination of hydrogen, which permit to Stancil et al. (1996) to develop a model of lithium primordial chemistry in the early Universe. Following these authors, one can say that LiH molecules are formed mainly through:

$$LiH^+ + H \mapsto LiH + H^+ \quad \text{(exchange reaction, 44)}$$

$$Li + H^- \mapsto LiH + e^- \quad \text{(associative detachment, 45)}$$

These reactions are coupled with the following destructions:

$$LiH + h\nu \mapsto Li + H \text{ (photodissociation, 29)}$$

$$LiH + H^+ \mapsto LiH^+ + H \text{ (charge transfer, 47)}$$

$$LiH + H \mapsto Li + H_2 \text{ (exchange reaction, 48)}$$

$$LiH + H^+ \mapsto Li^+ + H_2 \text{ (exchange reaction, 49)}$$

$$LiH + H^+ \mapsto Li + H_2^+ \text{ (exchange reaction, 50)}$$

Recently Gianturco & Gori Giorgi (1996) have carried out a fully quantum mechanical treatment of the reaction:

$$Li(2p) + H(1s) \mapsto LiH + h\nu \quad (51)$$

They have found that the radiative association of LiH molecule starting from lithium atoms that are electronically excited, Li(2p), is much more efficient that the one starting from lithium atoms in their ground state, Li(2s). The rate coefficients of this reaction is six order of magnitude greater than the one of reaction (28). The problem concerns the possibility to find cosmological epochs and scenarios for which

electronically excited lithium exist. This question deserves some comments depending on the cosmological environments and on the evolution of the primordial chemistry:

- Just after the short period in the expansion of the Universe referred to as the epoch of recombination
- Later in the evolution of the Universe, during the phase of the gravitational collapse of protostructure.

Let us briefly consider these two different chemical evolutions:

2.1. Molecular evolution after recombination

As it was already pointed out by Lepp & Shull (1984) LiH formation through radiative association (reaction 28) becomes effective when the photodestruction (reaction 29) decreases, i.e. for $z < 400$. On the other hand, to obtain, at these redshifts, the first electronic excitation of lithium atoms:

$$\mathrm{Li(2s)} + \mathrm{h}\nu \mapsto \mathrm{Li(2p)} \quad (52)$$

photons (with $\lambda \sim 0.69\ \mu\mathrm{m}$ or $E_\nu \sim 1.75\ \mathrm{eV}$) are needed. Since the de-excitation of lithium atoms through spontaneous emission

$$\mathrm{Li(2p)} \mapsto \mathrm{Li(2s)} + \mathrm{h}\nu \quad (53)$$

occurs in times of the order of 10^{-6} sec; a bath of such photons ($\lambda \sim 0.69\ \mu\mathrm{m}$, $E_\nu \sim 1.75\ \mathrm{eV}$) would have maintain lithium atoms in the first electronically excited state Li(2p). Gori Giorgi argues that (private communication 1996) if this background of photons is a black body ($\lambda \sim 0.69\ \mu\mathrm{m}$, $T \sim 3020$ K) the abundances of lithium atoms would have been such as:

$$\mathrm{Li(2p)} \sim 10^{-3}\mathrm{Li(2s)}$$

because the rate constant of reaction (51) is 10^6 greater than this of reaction (28) the final LiH abundance is more important. Moreover, one must also note that photons (with $\lambda < 0.50\ \mu\mathrm{m}$, $E_\nu > 2.5\ \mathrm{eV}$) could destroy LiH through photodissociation. Therefore Gori Giorgi concludes that the best opportunity would be the existence of a background of photons such as:

$$0.50\ \mu\mathrm{m} < \lambda < 0.69\ \mu\mathrm{m}$$

or

$$1.70\ \mathrm{eV} < E_\nu < 2.25\ \mathrm{eV}$$

What are the possible scenarios which could have led, at these redshifts $z < 400$, to a background of photons? One can already said that, at these redshifts: the existence of the first "light" from primordial objects (VMO, SMO ...) is a hypothetical possibility, the existence of a line at $\lambda \sim 0.69\ \mu\mathrm{m}$ is very difficult to imagine, but the existence of particle decay could be a plausible scenario.

2.2. Molecular evolution during the gravitational collapse of protostructures

In Puy & Signore (1996), we have seen that, in the framework of our very simple model, the collapse of a protocloud with a mass of $10^9\ M_\odot$ or $10^{10}\ M_\odot$ took place at a redshift of $z \sim 55$ or 25. At these redshifts, a primordial object (population III star for example) could have emitted photons in the interval [1.75 eV, 2.25 eV]. On the other hand, let us introduce the following reaction:

$$\mathrm{He} + \mathrm{H}^+ \mapsto \mathrm{HeH}^+ + \mathrm{h}\nu$$

which releases a photon at 1.85 eV. In the homogeneous case of the expansion of the Universe this reaction was only effective around a redshift of $z \sim 2000$ and therefore too earlier to excitate the lithium atoms.

But, during the gravitational collapse of protoclouds (at redshifts $z \sim 25$–55) while the density increased, the reaction (51) could have been effective. Therefore one can suppose that, during the collapse, the main part of lithium atoms Li(2s) could have been excited into Li(2p) through the reaction:

$$\mathrm{Li(2s)} + \mathrm{h}\nu(1.85\ \mathrm{eV}) \mapsto \mathrm{Li(2p)} \quad (52)$$

leading to the reaction:

$Li(2p) + H \mapsto LiH + h\nu$ (51)

which is more rapid than the reaction:

$Li(2s) + H \mapsto LiH + h\nu$ (28)

In the following and, in particular, in the new network, we will introduce two opposite cases for LiH formation:

- CASE 1: All lithium atoms are Li(2s); LiH formation is mainly given by reaction (28).
- CASE 2: All lithium atoms are Li(2p); LiH formation is mainly given by reaction (51) with its rate coefficient which is six orders of magnitude greater than the ones of reaction (28).

3. Results and discussions

We have chosen the same model of gravitational collapse developed in Puy & Signore (1996) from Lahav (1986). The evolution equations are described in this paper, these equations for a collapsing cloud, characterized by the mass M, are:

(conservation of matter)

$$\bullet \quad \frac{dn}{dt} = -3\frac{n}{r}\cdot\frac{dr}{dt} + \left(\frac{dn}{dt}\right)_{\text{chem}}$$

(evolution of the temperature T_m of the collapsing cloud)

$$\bullet \quad \frac{dT_m}{dt} = -2\frac{T_m}{r}\cdot\frac{dr}{dt} + \frac{2}{3nk}\cdot(\Xi_{\text{molec}} + \Theta_{\text{chem}})$$

(evolution of the radius r)

$$\bullet \quad \frac{d^2r}{dt} = \frac{5k}{\mu m_{\text{H}}}\cdot\frac{T_m}{r} - \frac{GM}{r^2}$$

where n is the density, t the time, $(\frac{dn}{dt})_{\text{chem}}$ the chemical contribution, k the Boltzmann constant, Ξ_{molec} the thermal function due to the primordial molecules H_2, HD and LiH, Θ_{chem} the thermal chemical function due to the enthalpy of the reactions, μ the mean molecular weight, m_{H} the mass of hydrogen and G the gravitational constant. All these equations are coupled with the equations of the chemical kinetic due to the chemical network given in Table 1. The calculations of the thermal function (molecular and chemical) are well-described in Puy & Signore (1997) which deals with the thermal analysis of this gravitational collapse. We recall that we consider a homologous collapse and the beginning of the collapse (i.e. for times lower than free-fall time), thus we neglect the effects of possible shock waves.

We consider protoclouds of masses $M = 10^{10} M_\odot$ and $M = 10^{11} M_\odot$. Table 2 gives the dynamical initial conditions at the turn-around redshift i.e. at the beginning of the collapse (see Padmanabhan (1993), and also Puy & Signore (1996), (1997)). Table 3 shows the initial abundances at the beginning of the collapse phase.

As in Puy & Signore (1996), (1997), we have chosen the initial abundances predicted by Inhomogeneous Big Bang Nucleosynthesis (IBBN).

3.1. $10^{10} M_\odot$ protocloud

Fig. 1 describe, at the beginning of the collapse ($t < t_{ff}$) the dynamical and chemical evolution of a $10^{10} M_\odot$ protocloud. For this mass the turn-around redshift z_{ta} is at 25 K (see Tables 2,3).

The abundances are calculated in the framework of non-excited lithium (Stancil et al. (1996), Case 1), the initial abundances are given in Table 3. We note (Fig. 1c) that the abundance of LiH decreases slightly mainly due to the destructions:

$LiH + H^+ \mapsto LiH^+ + H$ (charge transfer, 47)

$LiH + H \mapsto Li + H_2$ (exchange reaction, 48)

Fig. 2 represent the evolution of the abundance in the framework of excited lithium (Gianturco & Gori Giorgi (1996), Case 2). We note (Fig. 2c) that the situation is very different than the Case 1 (non-excited lithiums), the abundance increases significantly which shows the importance of the associative reaction with excited lithium.

In Figs. 1,2 the evolution of the abundances of H_2

Table 1
Table of chemical reactions

	Reactions	γ	α	β	ΔH	References
(1)	$H + e \rightarrow H^- + h\nu$	3×10^{-16}	1	0		Prasad and Huntress (1980)
(2)	$H^- + H \rightarrow H_2 + e$	1.5×10^{-9}	0	0	3.71	Launay et al. (1991)
(3)	$H^+ + H \rightarrow H_2^+ + h\nu$	1.8×10^{-18}	1.5	0		Dalgarno and Lepp (1987)
(4)	$H_2^+ + H \rightarrow H_2 + H^+$	6.4×10^{-10}	0	0	1.81	Karpas et al. (1979)
(5)	$H_2 + h\nu \rightarrow H + H$					Latter and Black (1991)
(6)	$H^- + h\nu \rightarrow H + e$					Wishart (1979)
(7)	$H_2^+ + h\nu \rightarrow H^+ + H$					Von Bush and Dunn (1972)
(8)	$H^+ + H^- \rightarrow H + H$	2.3×10^{-7}	-0.5	0	12.84	Prasad and Huntress (1980)
(9)	$H^+ + H^- \rightarrow H_2^+ + e$	8.83×10^{-15}	-0.32	0	1.9	Poulaerts et al. (1978)
(10)	$H_2^+ + H^- \rightarrow H_2 + H$	2.3×10^{-7}	-0.5	0	14.66	Prasad and Huntress (1980)
(11)	$H^+ + e \rightarrow H + h\nu$	3.5×10^{-12}	-0.70	0		Prasad and Huntress (1980)
(12)	$H_2^+ + e \rightarrow H + H$	1.68×10^{-8}	-0.29	0	10.95	Nakashima et al. (1987)
(13)	$H^+ + D \rightarrow D^+ + H$	$\times 10^{-9}$	0	41	-0.01	Watson (1976)
(14)	$H + D \rightarrow HD + h\nu$	$\times 10^{-25}$	0	0		Lepp and Shull (1984)
(15)	$D^+ + H_2 \rightarrow H^+ + HD$	1.7×10^{-9}	0	0	0.06	Smith et al. (1982)
(16)	$D^+ + H \rightarrow HD^+ + h\nu$	1.8×10^{-18}	1.5	0		Dalgarno and Lepp (1987)
(17)	$H^+ + D \rightarrow HD^+ + h\nu$	1.8×10^{-18}	1.5	0		Dalgarno and Lepp (1987)
(18)	$HD^+ + H \rightarrow H^+ + HD$	6.4×10^{-10}	0	0	1.85	Karpas et al. (1979)
(19)	$HD + h\nu \rightarrow H + D$					Latter and Black (1991)
(20)	$HD^+ + h\nu \rightarrow H^+ + D$					Von Bush and Dunn (1972)
(21)	$HD^+ + h\nu \rightarrow D^+ + H$					Von Bush and Dunn (1972)
(22)	$D^+ + e \rightarrow D + h\nu$	3.5×10^{-12}	-0.70	0		Prasad and Huntress (1980)
(23)	$HD^+ + e \rightarrow D + H$	1.39×10^{-8}	-0.29	0	10.93	Nakashima et al. (1987)
(24)	$D^+ + H \rightarrow H^+ + D$	$\times 10^{-9}$	0	0	0.01	Watson (1976)
(25)	$H^+ + HD \rightarrow D^+ + H_2$	1.7×10^{-10}	0	462	-0.06	Smith et al. (1982)
(26)	$H_2^+ + D \rightarrow D^+ + H_2$	6.4×10^{-10}	0	0	1.81	Karpas et al. (1979)
(27)	$D + H_2 \rightarrow HD + H$	1.09×10^{-18}	0	0	0.05	Zhang and Miller (1989)
(28)	$Li + H \rightarrow LiH + h\nu$	3.8×10^{-20}	0	0		Stancil et al. (1996)
(29)	$LiH + h\nu \rightarrow Li + H$					Kirby and Dalgarno (1978)
(30)	$Li^+ + e^- \rightarrow Li + \nu$	3.01×10^{-12}	-0.68	0		Stancil et al. (1996)
(31)	$Li + \nu \rightarrow Li^+ + e^-$					see Stancil et al. (1996)
(32)	$Li^+ + H^- \rightarrow Li + H$	1.2×10^{-7}	-0.36	$-16\,500$		Stancil et al. (1996)
(33)	$Li + H^+ \rightarrow Li^+ + H$	4.0×10^{-20}	6.8	1800		Stancil et al. (1996)
(34)	$Li + H^+ \rightarrow Li^+ + H + \nu$	2.3×10^{-14}	0.55	10 000		Stancil et al. (1996)
(34)	$Li + e^- \rightarrow Li^- + \nu$	1.85×10^{-15}	-0.62	9300		Stancil et al. (1996)
(35)	$Li^- + \nu \rightarrow Li + e^-$					see Stancil et al. (1996)
(36)	$Li^- + H^+ \rightarrow Li + H$	1.2×10^{-7}	-0.36	$-16\,500$		Stancil et al. (1996)
(37)	$Li^+ + H \rightarrow LiH^+ + \nu$	8.0×10^{-23}	-0.9	7000		Stancil et al. (1996)
(38)	$LiH^+ + \nu \rightarrow Li^+ + H$					see Stancil et al. (1996)
(39)	$Li + H^+ \rightarrow LiH^+ + \nu$	3.25×10^{-15}	-0.49			Stancil et al. (1996)
(40)	$LiH^+ + \nu \rightarrow Li + H^+$					see Stancil et al. (1996)
(41)	$LiH^+ + e^- \rightarrow Li + H$	2.6×10^{-8}	-0.47			Stancil et al. (1996)
(42)	$LiH^+ + H \rightarrow Li + H_2^+$	9.0×10^{-10}		66 400		Stancil et al. (1996)
(43)	$LiH^+ + H \rightarrow Li^+ + H_2$	3.0×10^{-10}				Stancil et al. (1996)
(44)	$LiH^+ + H \rightarrow LiH + H^+$	1.0×10^{-11}		67 900		Stancil et al. (1996)
(45)	$Li + H^- \rightarrow LiH + e^-$	4.0×10^{-10}				Stancil et al. (1996)
(46)	$Li^- + H \rightarrow LiH + e^-$	4.0×10^{-10}				Stancil et al. (1996)
(47)	$LiH + H^+ \rightarrow LiH^+ + H$	1.0×10^{-9}				Stancil et al. (1996)
(48)	$LiH + H \rightarrow Li + H_2$	2.0×10^{-11}				Stancil et al. (1996)
(49)	$LiH + H^+ \rightarrow Li^+ + H_2$	1.0×10^{-9}				Stancil et al. (1996)
(50)	$LiH + H^+ \rightarrow Li + H_2^+$	1.0×10^{-9}				Stancil et al. (1996)

The reaction rates are $k = \gamma(\frac{T}{300})^{\alpha}\exp(\frac{-\beta}{T})$, where γ is in $cm^3\,s^{-1}$, β in Kelvin. The enthalpy ΔH is in eV.

Table 2
Dynamical parameters at the beginning of the collapse

M ($M_\odot$)	z_{ta}	T_r (K)	T_{init} (K)	n_{init} ($10^5\,cm^{-3}$)	r_{ta} (10^{20} cm)
10^{11}	11	33	6	1.23×10^{-4}	633.2
10^{10}	25	72	25	1.23×10^{-3}	136.4

Rotation and magnetic field are neglected. z_{ta} is the turn-around redshift. T_r the temperature of the radiation at z_{ta}. T_{init} the initial temperature of the collapse. n_{init} and r_{init} respectively the initial density and initial radius of the collapsing protocloud.

Table 3
Initial relative abundances for $10^{10}M_\odot$ and $10^{11}M_\odot$ protoclouds

M ($M_\odot$)	z_{ta}	H_2 ($10^{-6}\,cm^{-3}$)	HD ($10^{-8}\,cm^{-3}$)	LiH ($10^{-17}\,cm^{-3}$)	H^+ ($10^{-5}\,cm^{-3}$)
10^{11}	11	8.4	5.21	0.3	3
10^{10}	25	8.4	5.21	0.3	3

At $z = z_{ta}$.

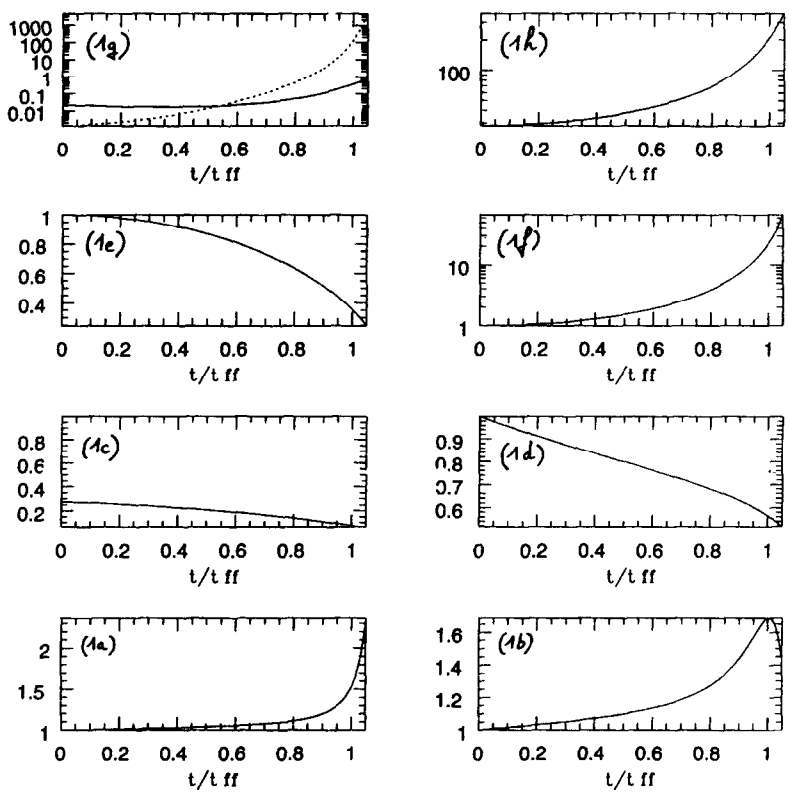


Fig. 1. Dynamical and chemical evolution of collapsing protocloud $M = 10^{10}M_\odot$ in the framework of non-excited lithium (Case 1). (1a) represents the evolution of H_2 molecule abundance normalised to the abundance of H_2 at z_{ta} the turn-around redshift. (1b) represents the evolution of HD abundance normalised to the abundance of HD at z_{ta}. (1c) characterizes the evolution of LiH abundance normalised to $10^{-17}\,cm^{-3}$. (1d) represents the evolution of H^+ normalised to the abundance of H^+ at z_{ta}. The series of curves (1e), (1f), (1g), (1h) represents respectively the evolution of radius and density normalised to their respective value at Z_{ta}, cooling (dotted line) and heating (solid line) normalised to $10^{30}\,erg\,cm^{-3}\,s^{-1}$, temperature.

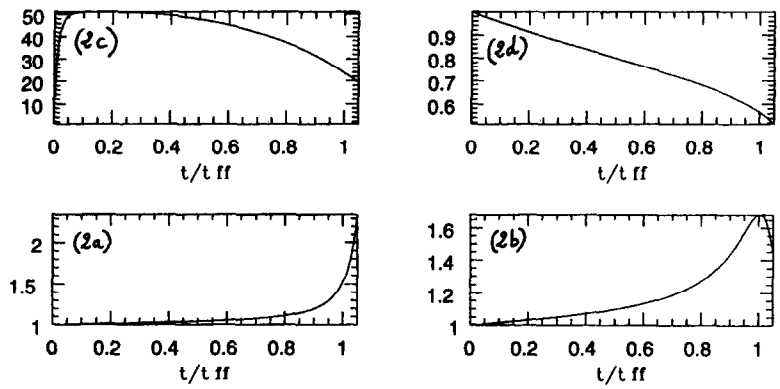


Fig. 2. Dynamical and chemical evolution of collapsing protocloud $M = 10^{10} M_{\odot}$ in the framework of excited lithium (Case 2, Gianturco & Gori Giorgi, 1996). (2a) represents the evolution of H_2 molecule abundance normalised to the abundance of H_2 at z_{ta} the turn-around redshift. (2b) represents the evolution of HD abundance normalised to the abundance of HD at z_{ta}. (2c) characterizes the evolution of LiH abundance normalised to $10^{-17}\,\mathrm{cm}^{-3}$. (2d) represents the evolution of H^+ normalised to the abundance of H^+ at z_{ta}.

and HD are similar than in our precedent article (Puy & Signore, 1996), the abundances of species having lithium are too low to participate to the modification of the abundances of H_2 or HD. In these two cases (Case 1 and Case 2) the dynamical evolution, characterized by Fig. 1e–Fig. 1h is similar. The abundance of lithium hydride is too low to affect the dynamic through the thermal function (cooling or heating) produced by LiH molecules. As we have seen in Puy & Signore (1997), the main cooling agent is the HD molecules (Fig. 1g).

3.2. $10^{11} M_{\odot}$ protocloud

In this case, we have only considered the case with excited lithium (Case 2). We note (Fig. 3c) a similar evolution for the abundance of LiH in the sense that LiH abundance fastly increases then decreases. Nevertheless the abundance of LiH have globally increased.

Concerning the dynamical evolution, the situation is close to $10^{10} M_{\odot}$. At the turn-around redshift, the temperature of the background radiation is greater than the temperature of matter inside the collapse. Thus at the beginning of the collapse, the molecules heat the medium. Then the temperature increases and reaches the value of the temperature of the background radiation, leading to the efficiency of the molecular cooling (dotted line in Fig. 3g) mainly due to HD molecules.

Finally, we note in this study the importance of the radiative association to form the LiH molecules. Moreover the exchange reactions (47), (48), (49) and (50) play an important role concerning the ways of destruction of LiH. Like Stancil et al. (1996) have mentioned, these gas-phase reactions and particularly

$$LiH + H \mapsto Li + H_2 \quad \text{(exchange reaction, 48)}$$

which reduces the abundance of LiH, the rate coefficients are uncertain and probably too large but not unreasonable (Dalgarno (1995), private communication). Nevertheless the trend concerning the evolution of LiH abundance inside the collapse is not clear (increases or decreases?) and more importantly the percentage of primeval lithium converted to LiH is quite uncertain (before and during the collapse phase). Therefore the importance of the role of the LiH molecule as a tracer of protostructure evolution cannot be definitively determined.

For all these reasons an important work remains to do concerning the specific calculations of the re-

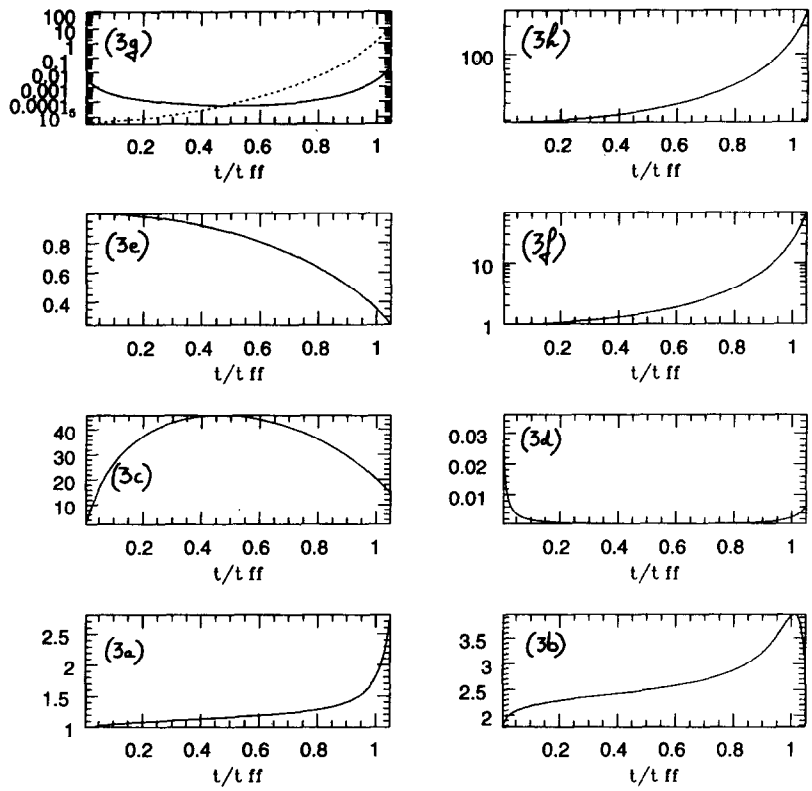


Fig. 3. Dynamical and chemical evolution of collapsing protocloud $M = 10^{11} M_{\odot}$ in the framework of excited lithium. (3a) represents the evolution of H_2 molecule abundance normalised to the abundance of H_2 at z_{ta} the turn-around redshift. (3b) represents the evolution of HD abundance normalised to the abundance of HD at z_{ta}. (3c) characterizes the evolution of LiH abundance normalised to $10^{-17}\,\mathrm{cm}^{-3}$. (3d) represents the evolution of H^+ normalised to $10^{-3}\,\mathrm{cm}^{-3}$. The series of curves (3e), (3f), (3g), (3h) represents respectively the evolution of radius and density normalised to their respective value at z_{ta}, cooling (dotted line) and heating (solid line) normalised to $10^{30}\,\mathrm{erg\,cm}^{-3}\,\mathrm{s}^{-1}$, temperature.

action rates, and particularly the knowledge of the complete chemical network of LiH in the excited states: LiH($A^1\Sigma^+$) and LiH($B^1\Pi$).

Acknowledgements

The authors gratefully acknowledge Alexander Dalgarno, Paula Gori Giorgi, Bianca and Francesco Melchiorri for valuable discussions on this field. Part of the work of D. Puy has been conducted under the auspices of the *Dr Tomalla Foundation.*

References

COBRAS/SAMBA presentation of Assessment Study results ESA-SCI (1994).

Copi, C.J., Schramm, D.N., & Turner, M.S., 1995, Sci, 267, 192.

Dalgarno, A. & Lepp, S., 1987, in: Astrochemistry, eds. M.S. Vardya & S.P. Tarafdar (Dordrecht, Reidel), pp. 109–120.

Dalgarno A., Kirby, K., & Stancil, P.C., 1996, ApJ, 458, 397.

De Bernardis, P., et al., 1993, A&A, 269, 1.

De Bernardis, P., et al., 1994, in: Present and Future of the Cosmic Microwave Background, eds. Sanz et al. (Berlin, Springer Verlag), pp. 188–207.

Dubrovich, V.K., 1993, AstL, 19, 53.

Gianturco, F.A. & Gori Giorgi, P., 1996, PhRvA, 54, 1.

Karpas, Z., Anicich, W.T., & Huntress, V.G., 1979, J. Chem. Phys. 70, 2877.
Kirby, K. & Dalgarno, A., 1978, ApJ, 224, 444.
Lahav, O., 1986, MNRAS, 220, 259.
Latter, W.B. & Black, J.H., 1991, ApJ, 372, 161.
Launay, J.M., Ledourneuf, M., & Zeippen, C.S., 1991, A&A, 252, 842.
Lepp, S. & Shull, M.J., 1984, ApJ, 280, 465.
Maoli, R., et al. 1994, ApJ, 425, 372.
Maoli, R., et al. 1996, ApJ, 457, 1.
Melchiorri, B. & Melchiorri, F., 1994, Riv. Nuovo Cimento 17, 1.
Nakashima, K., Takanagi, H., & Nakamura, H., 1987, J. Chem. Phys. 86, 726.
Padmanabhan, T., 1993, in: Structure Formation in the Universe (Cambridge, Cambridge University Press).
Prasad, S.S. & Huntress, W.T., 1980, ApJS, 43, 1.
Poulaerts, G., et al., 1978, J. Phys. B 11, L671.
Puy, D., et al. 1993, A&A, 267, 337.
Puy, D. & Signore, M., 1996, A&A 305, 371.
Puy, D. & Signore, M., 1997, NewA, 2, 299.
Sarkar, S., 1996, RPPh, 59, 1493.
Saslaw, W.C. & Zipoy, D., 1967, Natur, 216, 976.
Schramm, D.N. & Turner, M.S., 1997, Rev. Mod. Phys., in press.
Shchekinov, Y.A. & Entel, M.B., 1983, SvA, 27, 622.
Stancil, P.C., Lepp, S., & Dalgarno, A., 1996, ApJ, 458, 401.
Signore, M., et al. 1994, ApJS, 92, 535.
Signore, M., et al. 1997, Astron. Lett. Commun. 35, 349.
Smith, D., et al. 1982, ApJ, 263, 123.
Von Bush, F. & Dunn, G.H., 1972, PhRvA, 5, 1726.
Watson, W.D., 1976, Rev. Mod. Phys. 48, 513.
Wishart, A.W., 1979, MNRAS, 187, 59P.
Zhang, J.Z.H. & Miller, W.H., 1989, J. Chem. Phys. 91(3), 1528.